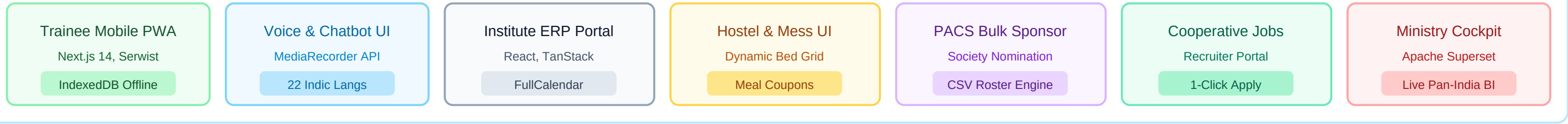


SAHAKAR-SETU: MASTER TECHNICAL SYSTEM ARCHITECTURE & DATA FLOW

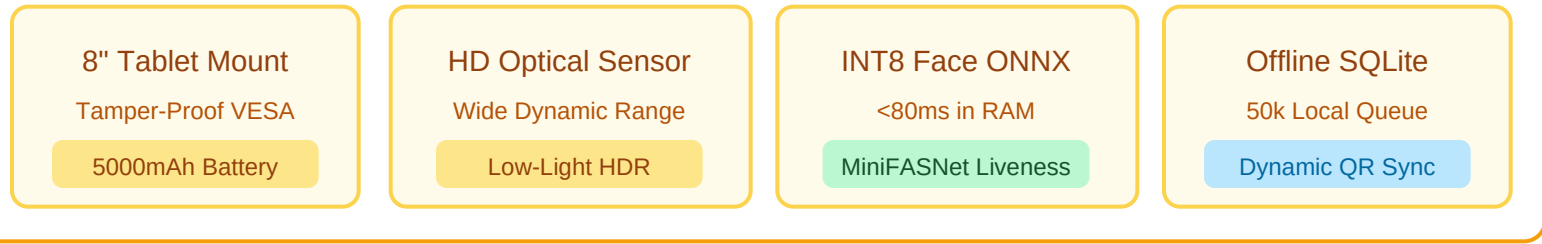
Ministry of Cooperation • National Council for Cooperative Training (NCCT) • 45 Integrated Components

LAYER 1: CLIENT ACCESS & PHYSICAL EDGE DEVICES

PANEL A: MOBILE & WEB CLIENTS



PANEL B: PHYSICAL HARDWARE KIOSK (CLASSROOM DOOR)



HTTPS / TLS 1.3: Voice Audio (WebM), Phone OTP, Profile Updates, Bulk CSV Rosters

HMAC-SHA256 Encrypted Sync: 128-d Face Vectors, KioskID, Timestamps (No Raw Photos)

LAYER 2: UNIFIED API GATEWAY & REVERSE PROXY

Nginx / Cloudflare Edge

TLS 1.3 Strict Cipher

WAF & Rate Limiting (Token Bucket)

JWT & RBAC Multi-Institute Filter

Internal Reverse Proxy Dispatch (/api/v1/auth, /erp, /lms, /jobs, /creds)

LAYER 3: CORE APPLICATION MODULAR MONOLITH (FastAPI High-Performance Async Backend)

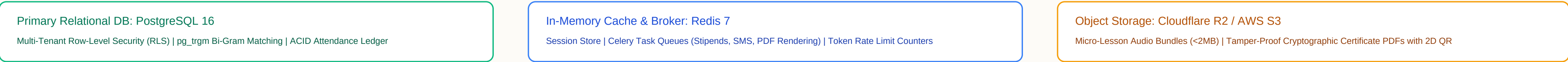


SQL with Row-Level Security (RLS) Tenant Isolation

Pub/Sub Event Bus & Celery Async Queues

Direct Presigned S3 URLs (Audio Chunks & Signed PDFs)

LAYER 4: PERSISTENCE & DATA STORAGE LAYER



gRPC Audio Streams (ASR & TTS)

JSON-LD W3C Verifiable Credential

REST: Bidirectional Vacancy Sync

REST: PACS Registration Verification

REST: DLT Notification Payload

LAYER 5: NATIONAL DIGITAL PUBLIC INFRASTRUCTURE (DPI) & EXTERNAL SERVICES



■ EDGE ATTENDANCE: <80ms Face Match | 100% Offline SQLite | Zero Raw Photos (DPDP Act)

■ VERIFICATION: Ed25519 2D QR | Sub-15ms Offline Validation | \$0 Blockchain Gas Fees

■ INFRASTRUCTURE: Modular Monolith | Scales to 20 Institutes & 79,630 PACS on <\$90/mo VPS